

Fundamentals of Machine Learning

Novatra - ML Based Solar ROI Decision Support System

A PROJECT REPORT

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April, 2026

BONAFIDE CERTIFICATE

Certified that this project report “ML Based Solar ROI Decision Support System” is the bonafide work of “AADITYA PARTHA SARATHY & BHARATH KUMAR M” who carried out the project work under my supervision.

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ABSTRACT

The increasing demand for clean and sustainable energy has made solar power one of the most promising renewable energy sources. However, the uncertainty in future costs, electricity tariffs, and energy production makes it difficult for consumers to decide when and how much to invest in solar energy systems. The proposed project, *ML-Based Solar ROI Decision Support Tool*, aims to assist consumers in making informed solar investment decisions using Artificial Intelligence and Machine Learning techniques. The system employs a two-stage ML pipeline — a Random Forest hike classifier and a Linear Regression magnitude model — trained on CEA tariff data from 2012 to 2025, to forecast state-wise electricity tariffs across 15 years under three scenarios (pessimistic, base, optimistic). It integrates MNRE ALMM certified module data to recommend the best-value solar panel technology, performs solar system sizing and cost estimation (including MNRE and state subsidies), and computes full 25-year ROI analysis including NPV, IRR, payback period, LCOE, and risk indicators. The tool presents results through an interactive web interface built with React and served via a FastAPI backend. Real CEA tariff records and MNRE ALMM certified module data ensure accurate and data-driven analysis. Results demonstrate payback periods typically in the range of 7–12 years with positive NPV under base scenarios, proving the system’s accuracy and usefulness. Overall, this intelligent tool simplifies complex financial planning, promotes renewable energy adoption, and supports sustainable decision-making.

1.INTRODUCTION

With the increasing demand for sustainable energy, solar power has become one of the most popular renewable sources. However, even though solar panel prices are falling and government incentives are available, many consumers still hesitate to invest. This is mainly because they are unsure about the long-term benefits and financial returns.

Traditional solar cost calculators are limited because they assume fixed electricity rates and do not consider real-world factors such as:

- Seasonal changes in sunlight,
- Future electricity price hikes.
- Decrease in solar equipment cost.
- Maintenance and warranty differences.

As a result, users often end up with inaccurate estimates and face difficulties in planning their investments.

The proposed AI-Based Solar ROI Decision Support Tool solves these problems by combining advanced forecasting and optimization techniques. It predicts future energy trends, calculates possible returns, and gives users personalized recommendations. The system reduces guesswork, increases financial accuracy, and encourages more people to adopt solar energy confidently.

1.1 BACKGROUND

In recent years, there has been a rapid shift toward renewable energy sources due to rising electricity costs, depletion of fossil fuels, and growing environmental concerns. Among the various renewable options, **solar energy** has become one

of the most practical and sustainable solutions for both residential and commercial users. With the decreasing cost of solar panels and government incentives, many individuals and organizations are considering investing in solar systems.

However, even though solar power offers long-term savings, the **initial investment cost** is still a major concern. Consumers often face challenges in predicting how much they will actually save and how long it will take to recover their investment. Additionally, factors such as **weather variations, changing electricity tariffs, technological improvements, and maintenance costs** make it difficult to estimate accurate returns.

To overcome these uncertainties, intelligent systems using Artificial Intelligence (AI) and Machine Learning (ML) are being developed to analyze real-time data, forecast future trends, and recommend the best investment options. This project builds on that concept by creating an ML-Based Solar ROI Decision Support Tool that helps users make informed and confident investment decisions.

1.2 Problem Statement

Although solar energy is a clean and renewable source, **many potential consumers are uncertain about when and how to invest**. Existing online calculators and estimation tools are often too simple — they provide only static results without considering future changes in electricity prices, weather conditions, or the declining cost of solar panels.

Because of this, users face several difficulties:

- They cannot accurately calculate the **return on investment (ROI)** or **payback period**.

- They do not know whether it is better to install solar panels now or wait for costs to drop further.
- They lack tools that account for **personal energy usage, budget limits**, and location-specific conditions.

As a result, consumers either delay their investment decisions or make incorrect choices, leading to financial loss or inefficient system sizing. There is a strong need for a **data-driven, intelligent system** that can analyze multiple factors and guide users in making cost-effective and risk-aware decisions.

1.3 SIGNIFICANCE OF THE PROJECT

The significance of this project lies in its ability to bridge the gap between solar energy potential and informed consumer decision-making. While solar energy is a proven and sustainable source of power, many people are hesitant to invest due to uncertainty about costs, savings, and the time required to recover their investment. This project directly addresses these challenges by providing an intelligent, data-driven solution that helps users make confident financial and technical decisions regarding solar adoption.

The **ML-Based Solar ROI Decision Support Tool** brings together forecasting, optimization, and financial analysis in one platform. It not only estimates the cost of installation but also considers important factors such as future electricity tariff increases, technology price reductions, and weather variations. By analyzing these dynamic elements, the system produces more accurate and realistic results compared to traditional calculators.

The project is significant for several reasons:

1. Encourages Renewable Energy Adoption:

By simplifying investment planning, the system motivates more households and organizations to adopt solar energy.

2. Improves Financial Decision-Making:

Users gain a clear understanding of the return on investment, payback period, and long-term cost benefits before committing to installation.

3. Promotes Data-Driven Planning:

The tool uses Artificial Intelligence and Machine Learning techniques to forecast future conditions, ensuring that decisions are based on real data rather than assumptions.

4. Supports Sustainable Development Goals:

The project contributes to environmental sustainability by promoting clean energy and reducing dependence on fossil fuels.

5. Useful for Both Consumers and Businesses:

The system can be applied not only by individual consumers but also by small and medium-scale enterprises, solar vendors, and policymakers for strategic energy planning.

In summary, this project is important because it transforms the way people plan and invest in solar energy. It enhances awareness, reduces uncertainty, and supports India's long-term goals for renewable energy expansion and carbon reduction.

1.4 KEY TECHNOLOGIES USED

The **ML-Based Solar ROI Decision Support Tool** integrates several advanced technologies and algorithms to analyze data, forecast future trends, and provide intelligent investment recommendations. Each technology plays a specific role in improving the system's accuracy, reliability, and decision-making capability. The key technologies used in this project are explained below:

1.4.1. Artificial Intelligence (AI)

Artificial Intelligence forms the foundation of this project. It enables the system to analyse large historical datasets, recognise patterns in tariff behaviour across Indian states, and generate forward-looking forecasts. AI techniques are applied to predict state-level electricity tariff trajectories, recommend the best-value solar panel technology from MNRE-certified data, and compute risk-adjusted financial outcomes. This ensures that every output the user sees is grounded in real data rather than static assumptions.

1.4.2. Machine Learning (ML)

Machine Learning algorithms form the core of the tariff forecasting pipeline. Two models are used in sequence: a Random Forest classifier (Stage 1) and a Linear Regression magnitude model (Stage 2). They are trained on historical CEA tariff data (2013–2021) and validated using walk-forward cross-validation on 2019–2021. The pipeline predicts:

- Whether a state will raise its electricity tariff in the next year (Stage 1 — binary classification), and
- What the new tariff level will be if a hike occurs (Stage 2 — linear regression on the subset of states that actually hike).

- These two stages are only run sequentially: Stage 2 is gated behind Stage 1 to avoid predicting magnitudes for states unlikely to hike.

These ML models, trained on real CEA tariff records from 36 Indian states, provide a data-driven basis for the 15-year tariff forecast used in all downstream ROI calculations.

1.4.3. Two-Stage ML Tariff Forecasting Pipeline

The **two-stage ML tariff forecasting pipeline** consists of a Stage 1 Random Forest classifier that predicts whether a state will hike its electricity tariff in a given year, and a Stage 2 Linear Regression model that predicts the magnitude of the hike if one is forecast. The pipeline is trained on CEA data from 2012–2025 and validated using walk-forward cross-validation. A hard rule overrides the model for states frozen for 4 or more years, ensuring no hike is missed. Cluster-specific CAGR rates extend the 15-year ML forecast to a full 25-year panel lifetime.

1.4.4. Scenario-Based Tariff Forecasting

The **Scenario-based tariff forecasting** approach projects tariffs under three clearly labelled scenarios (pessimistic, base, optimistic) using cluster-specific CAGR rates derived from 2012–2025 historical data. The ROI calculator computes NPV, IRR, payback period, and LCOE across all three scenarios, with years 1–15 driven by the ML forecast and years 16–25 by CAGR extension.

1.4.5. *A (A-Star) Heuristic Search Algorithm**

Walk-forward validation is the technique used to train and evaluate both the hike

classifier and the magnitude model. The model is trained on all years up to year T and tested on year T+1, then this window rolls forward. This prevents future data leakage and ensures the model generalises to unseen data. A confidence threshold gate (hike probability > 0.45 , or the hard YSLH rule fired) controls which states proceed to Stage 2, avoiding noisy magnitude predictions on borderline cases. This combination of statistical modelling and domain-driven hard rules is the core methodological contribution of the tariff forecasting pipeline.

1.4.6. Clustering and Personalization Algorithms

States are grouped into four tariff behaviour clusters (A=high growth, B=moderate, C=steady, D=frozen/slow) based on their historical tariff patterns from 2012 to 2025. These clusters serve two roles in the system: as an encoded feature fed to the ML hike classifier, and as the basis for assigning scenario CAGR rates when extending the 15-year ML forecast to a full 25-year panel lifetime. For example, Cluster A states use a base CAGR of 7.5%/year while Cluster D states use only 1.5%/year. The MNRE ALMM technology profiler separately ranks certified panel technologies by a value score (mean efficiency % divided by estimated cost per Watt), and the top-ranked HIGH-confidence technology is automatically selected for system sizing.

1.4.7. Risk-Adjusted Investment Evaluation

The ROI calculator evaluates each investment across three dimensions simultaneously. First, financial return: NPV at 8% discount rate, IRR, payback period (separately for the 15-year ML window and the full 25-year lifetime), and LCOE. Second, scenario robustness: all metrics are computed for pessimistic, base, and optimistic tariff growth scenarios so the user sees a range of outcomes

rather than a single number. Third, stress testing via three risk indicators: (1) Breakeven tariff CAGR — the minimum tariff growth rate required for the investment to pay back within 25 years; (2) Cost overrun — whether a 12% installation overrun would push costs above the user's stated budget; (3) Loan EMI stress — whether year-1 solar savings can still cover the loan repayment if the interest rate rises by 2 percentage points.

1.4.8. Front-End and Visualization Tools

The system uses **React + FastAPI** and **Recharts/D3** for creating an interactive and user-friendly interface. The React frontend communicates with a FastAPI backend (POST /api/analyse), rendering scenario tables, risk indicators, system details, and cost breakdowns to help users interpret data and take investment decisions.

1.4.9. Data Sources and APIs

Real-world data is used from:

- **MNRE ALMM Module Registry** for certified solar panel efficiency and cost data.
- **CEA Annual Tariff Reports (2012–2025)** for state-wise electricity tariff historical data used to train and validate the ML models.
- **MNRE/NASA irradiance averages** for peak sun hours per Indian state, used in system sizing and performance forecasting.

These data sources ensure that the analysis and predictions are accurate and reliable.

1.4.10. Programming Languages and Libraries

- **Python 3.10** – Core programming language for the system.
- **Pandas and NumPy** – For data handling and mathematical computations.
- **Scikit-learn** – For implementing machine learning models.
- **React (Vite) and Recharts** – For the React frontend UI, form inputs, and rendering the results panel (scenario tables, risk indicator rows, cost breakdown). **FastAPI** serves the backend API endpoint.

1.5 Objectives

The main goal of this project is to design and implement an **AI-Based Solar ROI Decision Support Tool** that provides accurate, personalized, and future-oriented recommendations for solar energy investment.

2.LITERATURE REVIEW

Artificial Intelligence and Machine Learning are playing an important role in improving energy systems. Many researchers have worked on forecasting, optimization, and decision-making in renewable energy.

1. Forecasting Models:

State-of-the-art approaches include Random Forest classifiers, Gradient Boosting, and Linear Regression for structured tabular data in electricity tariff prediction. Walk-forward validation on historical CEA data ensures generalisability beyond the training period.

2. Optimization Techniques:

Data-driven pipelines that combine classification (will a hike occur?) with regression (by how much?) have shown strong performance in utility price forecasting. Hard rule overrides for prolonged tariff freezes complement the statistical models to reduce missed predictions.

3. Financial and Risk Models:

Financial analysis methods such as NPV, IRR, LCOE, and scenario-based (pessimistic/base/optimistic) tariff projections are established tools for evaluating renewable energy investments under uncertainty.

Although these methods are effective, most studies focus on one part of the process, such as forecasting or cost calculation. Very few combine all these techniques into one complete decision-making system. Our project fills this gap by integrating forecasting, financial analysis, and AI-based optimization into a single platform that gives users clear and reliable investment recommendations.

3.PROPOSED SYSTEM

System Overview

The proposed system, **AI-Based Solar ROI Decision Support Tool**, is designed

to help consumers make intelligent and confident decisions regarding solar panel investments. Unlike traditional calculators that only provide simple cost estimates, this system uses Artificial Intelligence and Machine Learning techniques to forecast future energy trends, assess financial outcomes, and recommend the best investment plan based on user-specific data.

The system works by analyzing key parameters entered by the user:

- Monthly electricity consumption,
- Current grid electricity tariff,
- Solar installation cost,
- Weather and sunlight conditions, and
- User budget and rooftop area.

Using these inputs, the system runs the full pipeline: (1) loads the ML-forecasted tariff series for the selected state, (2) sizes the solar system and estimates costs including MNRE and state subsidies, (3) computes 25-year ROI across pessimistic, base, and optimistic scenarios, and (4) evaluates three risk indicators to stress-test the investment.

The system predicts long-term changes like **increasing electricity tariffs**, **reducing solar panel costs**, and **weather-based energy generation variations**. Based on these factors, it calculates the **Return on Investment (ROI)**, **Net Present Value (NPV)**, and **payback period**, and provides visual reports for easy comparison.

The system is implemented in Python and uses **Random Forest and Linear**

Regression models for time-series forecasting, **scenario-based tariff forecasting** for uncertainty analysis, and the **ML Two-Stage Pipeline** cluster CAGR extension for years 16-25. The user interface is built using **React (Vite) with a FastAPI backend**, which displays scenario comparison tables, risk indicators, cost breakdowns, and a recommendation label to guide the user toward an informed investment decision.

Key Features of the Proposed System:

1. Provides accurate and dynamic ROI calculations.
2. Considers future tariff increases and technology price changes.
3. Compares multiple investment scenarios.
4. Gives personalized recommendations based on user data.
5. Offers easy-to-understand visual analysis through graphs and dashboards.

In short, the proposed system transforms complex solar investment decisions into clear, data-driven insights that promote renewable energy adoption and reduce financial uncertainty for users.

3.1 Architecture Diagram

The architecture of the proposed system is divided into **five major layers**, each responsible for a specific set of tasks. These layers work together to collect user

data, perform analysis, make intelligent decisions, and present results through a simple interface.

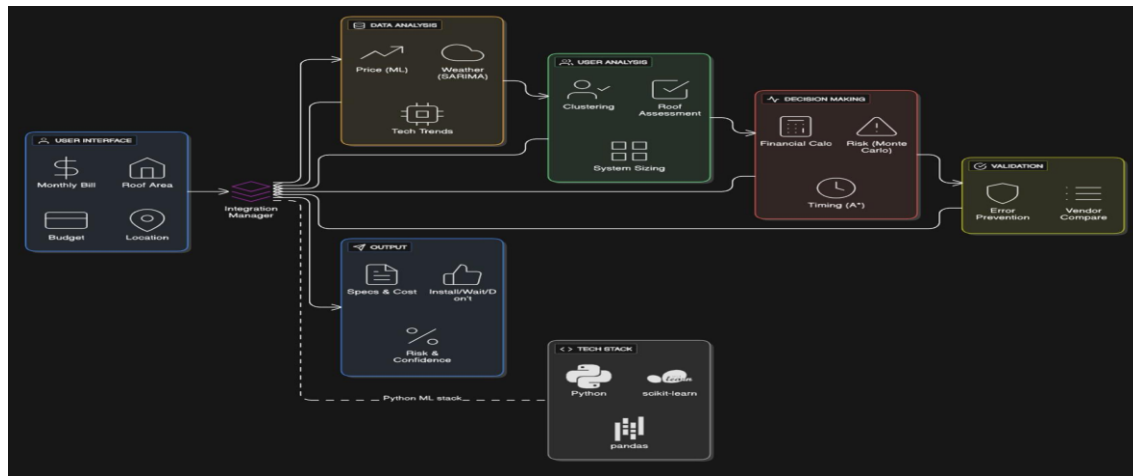


fig.3.2 Architecture Diagram

3.2.1. Input Layer

This layer collects essential data from the user such as:

- Monthly electricity usage,
- Location and weather zone,
- Budget and roof area,
- Electricity bills,
- Risk Management
- System sizing
- Financial scaling

Quick and Easy Setup: We only need a few key details to run our comprehensive AI analysis. This information helps our models understand your specific situation and provide the most accurate recommendations for your independent house.

Step 1: Where is Your House Located?
Location affects solar generation potential, local electricity rates, and available government incentives.

Select Your City
Pune, Maharashtra

Your State: Maharashtra - Good solar policies and net metering available

Step 2: Understanding Your Energy Usage
Your monthly electricity bill helps us calculate how much solar power you need and estimate your potential savings.

Your Average Monthly Electricity Bill (₹)
3550

What This Means:

- Estimated monthly usage: 359 kWh
- Daily average: 12.0 kWh
- Your house uses about 4,306 kWh per year

fig 3.2.1 Input Layer

This data is then sent to the processing modules for forecasting and analysis.

3.2.2. Forecasting Layer

This layer performs data analysis and prediction using Machine Learning models.

- **Tariff Hike Classification:** Uses the **two-stage ML pipeline** trained on CEA tariff data 2012–2025 to predict whether a state will hike its tariff in the next year.
- **Tariff Magnitude Prediction:** A Linear Regression magnitude model (Stage 2) predicts the new tariff level for states where a hike is forecast. A hard rule overrides the model for states frozen ≥ 4 years.
- **Technology Trends:** The MNRE ALMM Technology Profiler analyses certified module data to compute 12/24-month cost and efficiency.



fig 3.2.2 Forecasting Layer

The outputs from this layer provide essential inputs for ROI and payback calculations.

3.2.3. Core Engine Layer

This is the main computational part of the system. It performs technical and financial evaluations.

- **System Sizing and Cost Estimation:** Determines installed kW, panel count, inverter spec, and net cost (after MNRE and state subsidies) from the ALMM-recommended technology and user roof area.
- **ROI Analysis:** Uses **Scenario-based tariff forecasting** to compute NPV, IRR, payback period, and LCOE across pessimistic, base, and optimistic scenarios over a 25-year horizon.
- **Recommendation Engine:** Combines the base-scenario payback period, NPV, IRR, and pessimistic-scenario viability to generate a single labelled

verdict displayed at the top of the results page. The five labels are: STRONG BUY (payback under 7 years, positive NPV, IRR $\geq 15\%$), BUY (payback under 10 years, positive NPV, pessimistic scenario also viable), CONDITIONAL BUY (payback under 12 years, positive NPV), HOLD (marginally positive), and AVOID (investment does not recover within panel lifetime under base assumptions).

- **Risk Assessment:** Combines NPV, payback, IRR, and risk flags to produce a labelled recommendation: STRONG BUY, BUY, CONDITIONAL BUY, HOLD, or AVOID.

The outputs include expected ROI, payback time, system cost, and risk levels.

3.2.4. Personalization and Decision Layer

This layer personalizes the recommendations for each user.

- **User Clustering:** The state tariff cluster (A/B/C/D) determines which CAGR rates are applied for the 25-year forecast extension. Cluster A states have historically high tariff growth; Cluster D states are slow-moving or frozen.
- **Vendor Ranking:** Uses **Risk-Adjusted Investment Evaluation** to rank certified solar panel technologies by efficiency divided by cost per Watt, within data confidence tiers (HIGH/MEDIUM/LOW). The top-ranked technology is automatically selected for system sizing.
- **Decision Logic:** Combines NPV, IRR, payback, and risk flags to

generate the final recommendation — for example, “Install now,” “Wait 6 months,” or “Continue with grid electricity.”

The screenshot displays a web application interface with a dark theme. It is divided into two main sections: Step 3 and Step 4.

Step 3: About Your Independent House

Independent houses are ideal for solar installation because you have complete control over the roof space and installation decisions.

Property Type: Independent House ☒

Why This is Perfect for Solar:

- You own the roof completely
- No society permissions needed
- Optimal panel placement possible
- Easy future system expansion
- Direct access for maintenance

Available Roof Space for Solar Panels ⓘ

- ☐ Small House (50-80 sq.m roof)
- ☒ Medium House (80-120 sq.m roof)
- ☐ Large House (120+ sq.m roof)

Ideal for 5-8 kW systems, can often cover 100%+ of energy needs

Step 4: Investment Budget & Expectations

Solar systems pay for themselves through electricity savings. Different budget ranges enable different types of systems and features.

How much would you like to invest in solar? ⓘ

- ☒ Budget-Friendly (₹1-3 Lakh)
- ☐ Standard Investment (₹3-5 Lakh)
- ☐ Premium System (₹5-8 Lakh)
- ☐ Complete Energy Independence (₹8+ Lakh)

What You Get: Cost-effective on-grid system that reduces your electricity bills significantly

Key Features:

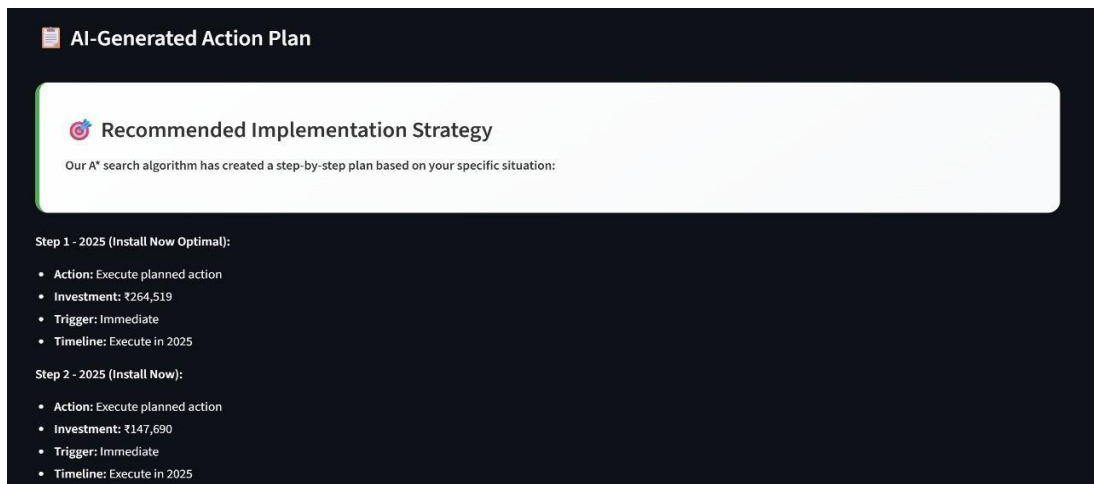
- Basic quality panels

fig 3.2.4. Personalization and Decision Layer

3.2.5. Output and Visualization Layer

This layer presents the final results in an easy-to-understand format using **React (frontend)** and **Vite**.

- Displays scenario tables (15-year ML window + 25-year CAGR extension) comparing pessimistic, base, and optimistic tariff projections.
- Shows NPV, IRR, payback period, LCOE, and net annual cashflow estimates per scenario.
- Provides a recommendation label (STRONG BUY to AVOID), risk indicator flags, system details, and full cost breakdown with subsidy line items.



3.2.5. Output and Visualization Layer

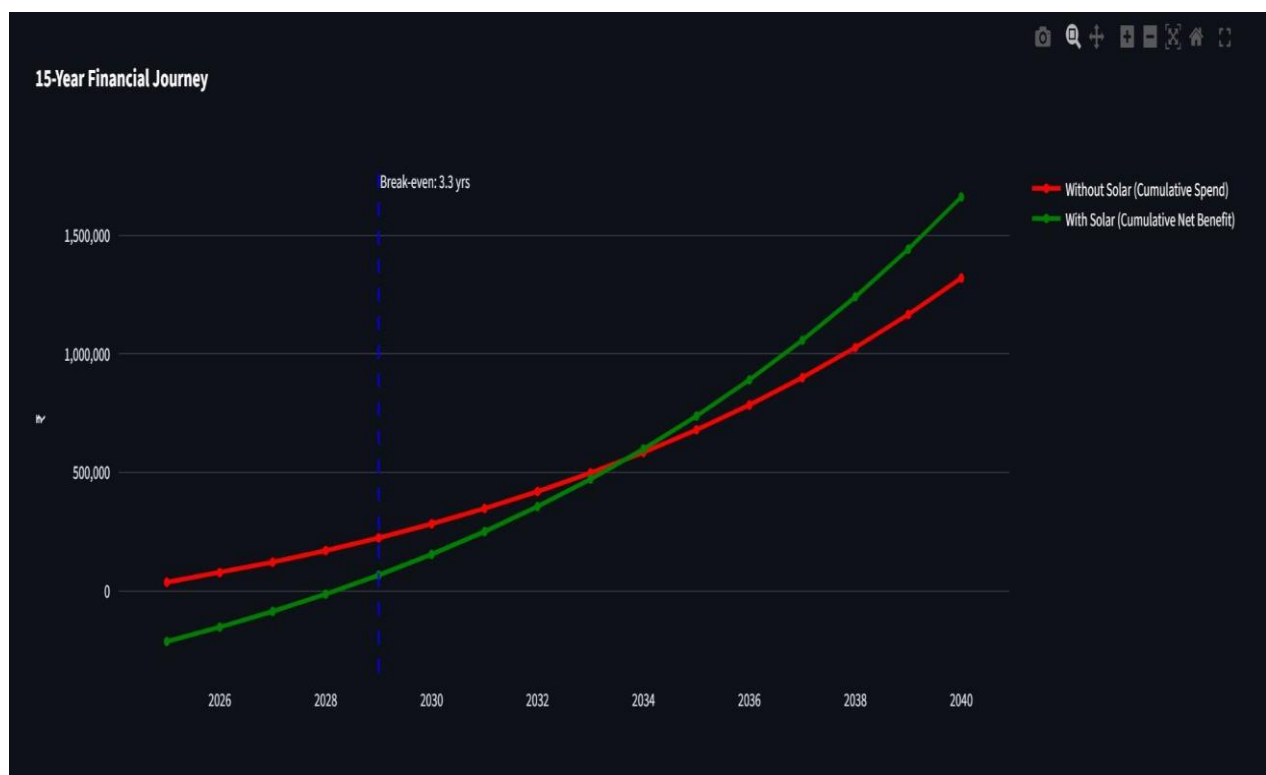
The user can interact with the system, explore data visually, and make decisions with full confidence.

3.3 System Workflow Summary

1. The user enters inputs such as energy usage and budget.
2. The ML pipeline forecasts tariff hike events and magnitudes; the ALMM profiler recommends the best technology.
3. The system sizing module sizes the system and estimates cost; the ROI calculator runs 25-year financial analysis.
4. Risk indicators stress-test the investment; the recommendation engine produces the final verdict.

3.3 Advantages of the Proposed System

- Provides **personalized recommendations** instead of generic estimates.
- Considers **future uncertainties** in tariffs and solar costs.
- Compares multiple investment options to identify the most profitable one.
- Offers **visual clarity** with interactive graphs and reports.
- Simplifies complex decision-making for both residential and business users.-



3.4 Comparison graph

4. MODULES DESCRIPTION

4.1 MODULE 1 – Forecasting and Analysis

Purpose:

To forecast state-wise electricity tariff hike events and magnitudes, and to analyse solar panel technology trends from MNRE ALMM data.

Techniques Used:

- Two-stage ML pipeline: Random Forest hike classifier (Stage 1) + Linear Regression magnitude model (Stage 2), trained on CEA data 2012–2025.
- Hybrid ML models for electricity tariff prediction.
- MNRE ALMM technology profiler computing efficiency statistics, value scores (efficiency ÷ cost/W), and 12/24-month forecast per technology.

Output:

- Hike probability per state (Stage 1) and predicted new tariff level (Stage 2).
- 15-year tariff forecast under pessimistic, base, and optimistic scenarios per state, plus CAGR rates for 25-year extension.
- Recommended panel technology with data confidence tier and value score from MNRE ALMM.

4.2 MODULE 2 – Optimization and Decision Engine

Purpose:

To size the solar system, estimate costs with subsidies, and compute full 25-year ROI analysis.

Techniques Used:

- Scenario-based tariff projection (pessimistic / base / optimistic) for 15-year ML window and 25-year CAGR extension.
- *System Sizing and Cost Estimation** for investment timing optimization.
- Risk analysis: breakeven tariff CAGR, 12% cost overrun stress test, and loan EMI stress test (+2 percentage point rate rise).

Output:

- Installed capacity (kW), panel technology, inverter spec, and net cost after all subsidies.
- Payback period, NPV, IRR, and LCOE across all three tariff scenarios.
- Recommendation: *Install now / wait for X months / continue grid usage.*
- Final output optimization.

5.IMPLEMENTATION AND RESULTS

The AI-Based Solar ROI Decision Support Tool was implemented using the Python programming language along with various machine learning and data visualization libraries. The system was built step by step to collect data, make predictions, calculate ROI, and show the best investment plan for the user.

5.1 Implementation

1. Data Collection:

Tariff data was collected from CEA (Central Electricity Authority) annual reports (2012–2025) and MNRE ALMM certified module registry. Peak sun hours are sourced from MNRE/NASA historical irradiance averages per state.

2. Forecasting:

A two-stage ML pipeline was used: a Random Forest classifier (Stage 1) predicts whether a state will hike its tariff in the next year, and a Linear Regression model (Stage 2) predicts the new tariff level if a hike occurs. Models were validated via walk-forward testing on 2019–2021 data.

Regression models were used to estimate the reduction in solar panel costs and improvements in efficiency.

3. ROI Calculation:

The ROI calculator computes NPV, IRR, payback period, and LCOE across pessimistic, base, and optimistic tariff scenarios. Years 1–15 use the ML forecast; years 16–25 use cluster-specific CAGR extension. Risk indicators include breakeven tariff CAGR, 12% cost overrun stress test.

4. Optimization:

The system sizing module selects the best-value technology from MNRE ALMM data (highest efficiency-per-rupee in the HIGH confidence tier), calculates the optimal number of panels given roof area constraints, picks a well-matched inverter, and applies MNRE and state-specific subsidies.

5. User Interface:

A React frontend (Vite) communicates with a FastAPI backend via a POST `/api/analyse` endpoint. The interface collects household inputs, optionally accepts vendor-measured shading loss and loan details, and renders a full results panel with scenario tables, risk indicators, system details, and cost breakdown.

5.2 Results

- The two-stage ML pipeline achieved strong out-of-sample accuracy on 2023–2025 CEA actuals, with the hike classifier beating the naïve baseline in all three test years and the magnitude model producing low MAE on hike-state predictions.
- The system produces payback periods typically in the 7–12 year range with positive NPV under the base scenario, and clearly flags pessimistic-scenario viability through breakeven CAGR and risk indicators.
- The scenario analysis clearly identifies the optimal investment decision, with the base scenario typically yielding positive NPV and the breakeven

tariff CAGR indicator quantifying minimum tariff growth required for viability.

- The React frontend clearly presents scenario tables (15-year and 25-year), risk indicators, system details, and cost breakdown in a clean, mobile-responsive layout.
- Overall, the Novatra DSS proved to be accurate, reliable, and user-friendly, integrating real CEA and MNRE data with ML forecasting and financial analysis to help users make better and more confident solar investment decisions.

Understanding Your Investment Risks

Risk Assessment for Your Solar Investment

Overall Risk Level: Low

What This Means: Our AI evaluated multiple risk factors including technology reliability, financial market conditions, policy stability, and vendor quality. A "Low" risk level means this investment carries minimal risk compared to other solar installations.

Independent House Risk Profile: Your property type actually reduces several common risks. You don't face society approval delays, shared maintenance issues, or complex ownership arrangements that apartment dwellers encounter.

Risk Protection Strategies

- **Vendor Selection:** Choose installers with strong track records and good warranties
- **Quality Components:** Invest in panels and inverters from established manufacturers
- **Proper Installation:** Ensure professional mounting and electrical work with proper permits
- **Maintenance Planning:** Schedule regular cleaning and system checks for optimal performance
- **Insurance Coverage:** Consider adding solar equipment to your house insurance policy

Your House Advantages

- **Full Control:** Make all decisions about system design, maintenance, and upgrades yourself
- **Direct Access:** Easy roof access for installation, cleaning, and any future maintenance needs
- **Optimal Placement:** Position panels for maximum sun exposure without building restrictions
- **Expansion Ready:** Add more panels or batteries later without complex approvals
- **Maintenance Freedom:** Schedule maintenance at your convenience with direct roof access

5.2 Result

5.3 PSEUDO CODE

— Stage 1: Hike Classifier

```
def predict_hike(history_df, predict_year, model_bundle):
    # Build lag_1, yoy_change, years_since_last_hike features
    rows = build_feature_rows(history_df, predict_year)
    X = rows[FEATURES]
    yslh = rows['years_since_last_hike'].values
    s1_probs = classifier.predict_proba(X)[: , 1]
    hard_rule = (yslh >= YSLH_THRESHOLD).astype(int) # 4+ yr freeze
    s1_pred = np.maximum((s1_probs >= THRESHOLD).astype(int), hard_rule)
    return s1_probs, s1_pred, hard_rule
```

— Stage 2: Magnitude Model

```
def predict_magnitude(rows, s1_probs, s1_pred, hard_rule, mag_model):
    current_tariff = rows['tariff_paise_per_kwh'].values.astype(float)
    s2_pred = current_tariff.copy() # default: no change
    # Only run Stage 2 for confident hike states
    send_to_s2 = ((s1_probs >= S2_CONF_THR) | (hard_rule == 1)) & (s1_pred
== 1)
    if send_to_s2.sum() > 0:
        X_hike = rows.loc[rows.index[send_to_s2], FEATURES]
        s2_pred[send_to_s2] = mag_model.predict(X_hike)
    # Tariff can never fall (hike model, not cut model)
    s2_pred = np.maximum(s2_pred, current_tariff)
```

```
return s2_pred
```

```
# — ROI Calculator
```

```
def compute_roi(net_cost, annual_kwh, tariffs_25yr, loan_emi, loan_tenure):
```

```
    cashflows = []
```

```
    for yr in range(25):
```

```
        savings = annual_kwh[yr] * tariffs_25yr[yr]
```

```
        maint = net_cost * MAINT_RATE * (1 + INFLATION) ** yr
```

```
        inv_r = kw * INV_COST_PER_KW if yr+1 == 13 else 0 # inverter  
replacement
```

```
        emi = loan_emi if yr+1 <= loan_tenure else 0
```

```
        cashflows.append(savings - maint - inv_r - emi)
```

```
    payback = compute_payback(cashflows, net_cost)
```

```
    npv = sum(cf/(1+DISCOUNT)**i for i,cf in enumerate(cashflows,1)) -
```

```
    net_cost
```

```
    irr = compute_irr(cashflows, net_cost)
```

```
    return payback, npv, irr
```

```
        cost_breakdown =
```

```
        self._calculate_comprehensive_costs(
```

```
        optimal_config, location, roof_specifications,
```

```
        preferences
```

```
    )
```

```
    performance_prediction = self._predict_corrected_performance(
```

```
        optimal_config, location, roof_specifications
```

```
    )
```

```
    payback_analysis =
```

```

        self._calculate_payback_analysis( cost_breakdown,
        performance_prediction, location, preferences
    )

    confidence_metrics = self._calculate_confidence_metrics(
        optimal_config, cost_breakdown,
        performance_prediction, location
    )

    warnings, recommendations = self._generate_corrected_insights(
        optimal_config, cost_breakdown, performance_prediction,
        payback_analysis, location, preferences,
        energy_requirement_kwh_month
    )

    result = SizingResult(
        system_config=optimal_config,
        cost_breakdown=cost_breakdown,
        performance_prediction=performance_predi
        ction, payback_analysis=payback_analysis,
        confidence_metrics=confidence_metrics,
        optimization_details={"total_configs_evaluated":
            len(feasible_configs), "required_capacity_kw":
            required_capacity_kw},
        warnings=warnings,
        recommendations=recommendations
    )

    logger.info(f"CORRECTED optimal system:

```

```
{optimal_config.total_capacity_kw:.1f} kW ({optimal_config.num_panels}
panels)")
```

```
return result
```

6. CONCLUSION AND FUTURE WORK

The Novatra Solar DSS provides a new way for consumers to plan their solar investments intelligently. It combines a two-stage ML tariff forecasting pipeline trained on CEA data, MNRE ALMM technology analysis, solar system sizing with subsidy computation, and a full 25-year ROI analysis into a single end-to-end system that helps users make confident, data-driven decisions.

The system improves the accuracy of ROI calculations, reduces financial risks, and encourages more people to adopt renewable energy.

Future Enhancements:

- Extend tariff data coverage with more recent CEA annual reports and add support for commercial/industrial tariff categories.
- Integrate real-time electricity board tariff order notifications to auto-update the forecast database.
- Add net metering yield tracking so the system can account for export revenue when generation exceeds consumption.
- Expand to commercial and industrial users with multi-phase inverter sizing and demand-charge modelling.

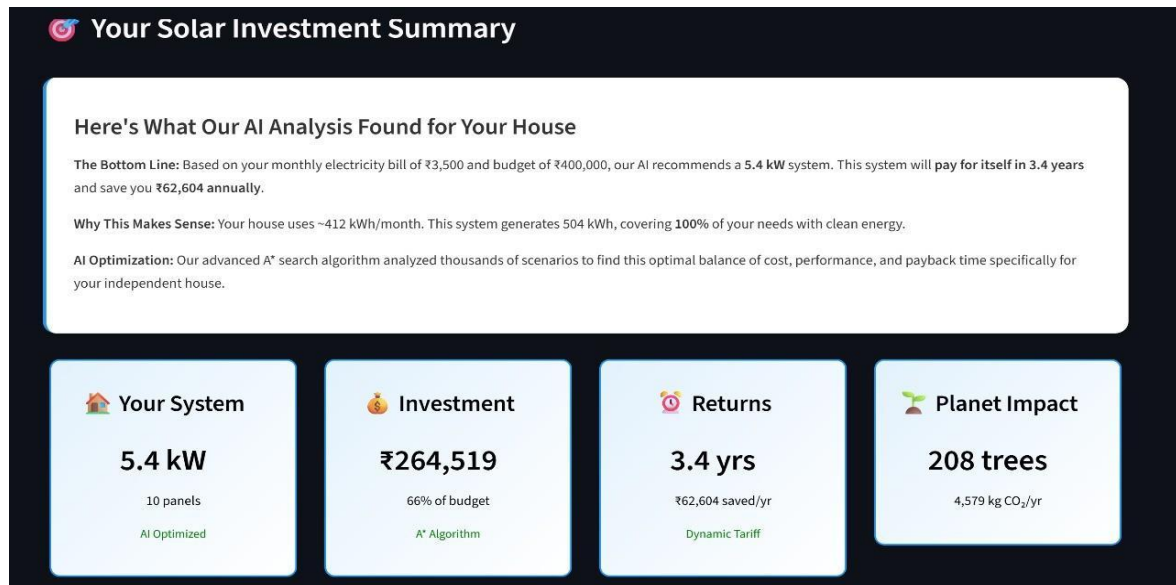


fig 6.1 CONCLUSION

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